

Mateus Freitas Paiva, PhD

Early-career researcher

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PROFILE SUMMARY

Early-career researcher in heterogeneous catalysis and sustainable lignocellulosic biomass valorization with international training in Brazil, France, and the Netherlands. PhD in Materials Chemistry from the Université de Lille (PEARL i-site / MSCA ecosystem) with European Doctorate Label. Currently a postdoctoral researcher at the Federal Fluminense University (UFF) and the National Institute of Technology (INT) –Brazil, and, supported by the Brazilian National Council for Scientific and Technological Development (CNPq). Strong publication record, peer-review activity, and experience mentoring students and leading interdisciplinary research projects, with active engagement in scientific outreach and research communication to promote accessible, evidence-based science.

SCIENTIFIC VISION

Commitment to building inclusive, collaborative research environments that integrate fundamental science, data-driven tools, and scientific communication – catalyzing both people and reactions to accelerate sustainable solutions and empower the next generation of scientists.

CURRENT POSITION

Postdoctoral Researcher – Federal Fluminense University (UFF) and National Institute of Technology (INT) – Brazil, (2024 – present)

- Leading an independently developed research agenda in catalytic biomass valorization, integrating experimental and data-driven strategies.

EDUCATION

PhD in Materials Chemistry – Université de Lille, France (2021–2024)

MSc in Chemistry (Catalysis) – University of Brasília (2019–2021)

BSc in Chemistry, with Technological Degree and Teaching Certification – University of Brasília (2013–2018)

EXPERIENCE

Junior researcher / Doctoral candidate – Université de Lille – CNRS, France (2021 – 2024)

- Designed and implemented integrated biorefinery-inspired strategies for lignocellulosic biomass fractionation; developed heterogeneous catalytic systems for cellulose, hemicellulose, and lignin valorization, including reductive catalytic depolymerization (RCD); conducted combustion and kinetic studies of bio-derived hydrocarbons; collaborated with international research teams and supervised junior researchers.

Student researcher (CNPq Fellow) – University of Brasília (UnB), Brazil (2014 – 2021)

- Synthesized and characterized heterogeneous catalysts (heteropolyacids, zeolites, mesoporous materials); performed catalytic reactions including esterification, sugar dehydration, polymerization, and biomass-derived transformations; served as teaching assistant (academic monitoring) for multiple undergraduate chemistry courses, supporting instruction, laboratory activities, and student learning.

MENTORSHIP & LEADERSHIP

Supervision and mentoring of undergraduate and graduate students in catalysis and biomass valorization projects; Training students in experimental design, data interpretation, and scientific communication; Leadership in collaborative and interdisciplinary research teams in Brazil and Europe; Team leader in international innovation and project-based training activities.

PUBLICATIONS

11 peer-reviewed journal articles, including publications in ChemSusChem, Cellulose, and Molecular Catalysis; 4 book chapters; and +38 contributions to national and international conferences. A complete publication list is available via the linked researcher profiles.

PEER REVIEW & SERVICE

Peer reviewer for international journals; Ad hoc consultant and evaluator for international catalysis conferences; Member of organizing committees for scientific meetings in catalysis and materials chemistry.

AWARDS & DISTINCTIONS

DivCat travel fellowship – Support to attend the 18th International congress on catalysis (ICC), French chemical society (SCF), France (2024)

Éole excellence fellowship – academic research stay in the Netherlands, Franco–Dutch higher education and research network (RFN), France (2021)

Dmitri Mendeleev award – Outstanding student award, Regional council of chemistry (CRQ-XII), Brazil (2021)

Best poster awards (Physical Chemistry and Catalysis Divisions) – 38th RA, 42nd RA and IV ERCO Meetings, Brazilian Chemical Society (SBQ), Brazil (2015, 2019 and 2021);

ICC Young scientists travel award – International Association of Catalysis Societies (IACS), United States (2020)

Outstanding scientific video award for research communication (Master's level) – Brazilian Catalysis Society (SBCat), Brazil, (2020)

Honorable mentions in exact sciences – Undergraduate research congresses, University of Brasília (UnB) (2015 and 2016).

TECHNICAL TRAINING

Formal training in leadership development, project management for research and innovation, and collaborative and interdisciplinary team dynamics; complementary advanced training in heterogeneous catalysis, biomass conversion and biorefinery processes, reaction kinetics and catalytic mechanisms, and advanced catalyst characterization techniques.

LANGUAGES

Portuguese – Native; English – Advanced; French – Advanced; Spanish – Intermediate.